

REFERENCE: SIMPLY-TYPED λ -CALCULUS

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Figure 1: Syntax chart of the little language of numbers and strings

types	τ	::=	Num		numbers
			Str		strings
pre-terms	e	::=	x		variables
			$\text{num}[n]$		numeral
			$\text{str}[s]$		string literals
			$\text{plus}(e_1; e_2)$	$e_1 + e_2$	addition
			$\text{times}(e_1; e_2)$	$e_1 * e_2$	multiplication
			$\text{cat}(e_1; e_2)$	$e_1 \mathbin{++} e_2$	concatenation
			$\text{len}(e)$	$ e $	length
			$\text{let}(e_1; x. e_2)$	$\text{let } x \leftarrow e_1 \text{ in } e_2$	let-definition

Figure 2: Statics of the little language of numbers and strings

$\frac{\text{VAR}}{\Gamma, x : \sigma \vdash x : \sigma}$	$\frac{\text{NUM} \quad n \in \mathbb{N}}{\Gamma \vdash \text{num}[n] : \text{Num}}$	$\frac{\text{STR} \quad s \in \Sigma^*}{\Gamma \vdash \text{str}[s] : \text{Str}}$
$\frac{\text{PLUS} \quad \Gamma \vdash e_1 : \text{Num} \quad \Gamma \vdash e_2 : \text{Num}}{\Gamma \vdash \text{plus}(e_1; e_2) : \text{Num}}$	$\frac{\text{TIMES} \quad \Gamma \vdash e_1 : \text{Num} \quad \Gamma \vdash e_2 : \text{Num}}{\Gamma \vdash \text{times}(e_1; e_2) : \text{Num}}$	
$\frac{\text{CAT} \quad \Gamma \vdash e_1 : \text{Str} \quad \Gamma \vdash e_2 : \text{Str}}{\Gamma \vdash \text{cat}(e_1; e_2) : \text{Str}}$	$\frac{\text{LEN} \quad \Gamma \vdash e : \text{Str}}{\Gamma \vdash \text{len}(e) : \text{Num}}$	$\frac{\text{LET} \quad \Gamma \vdash e_1 : \sigma_1 \quad \Gamma, x : \sigma_1 \vdash e_2 : \sigma_2}{\Gamma \vdash \text{let}(e_1; x. e_2) : \sigma_2}$

Figure 3: Dynamics of the little language of numbers and strings

$\frac{\text{VAL-NUM} \quad n \in \mathbb{N}}{\text{num}[n] \text{ val}}$	$\frac{\text{VAL-STR} \quad s \in \Sigma^*}{\text{str}[s] \text{ val}}$	$\frac{\text{D-PLUS} \quad n_1 + n_2 = n}{\text{plus}(\text{num}[n_1]; \text{num}[n_2]) \mapsto \text{num}[n]}$
$\frac{\text{D-PLUS-1} \quad e_1 \mapsto e'_1}{\text{plus}(e_1; e_2) \mapsto \text{plus}(e'_1; e_2)}$	$\frac{\text{D-PLUS-2} \quad e_1 \text{ val} \quad e_2 \mapsto e'_2}{\text{plus}(e_1; e_2) \mapsto \text{plus}(e_1; e'_2)}$	$\frac{\text{D-CAT} \quad s_1 \upharpoonright s_2 = s}{\text{cat}(\text{str}[s_1]; \text{str}[s_2]) \mapsto \text{str}[s]}$
$\frac{\text{D-CAT-1} \quad e_1 \mapsto e'_1}{\text{cat}(e_1; e_2) \mapsto \text{cat}(e'_1; e_2)}$	$\frac{\text{D-CAT-2} \quad e_1 \text{ val} \quad e_2 \mapsto e'_2}{\text{cat}(e_1; e_2) \mapsto \text{cat}(e_1; e'_2)}$	$\frac{\text{D-LEN} \quad s = n}{\text{len}(\text{str}[s]) \mapsto \text{num}[n]}$
$\frac{\text{D-LEN-1} \quad e \mapsto e'}{\text{len}(e) \mapsto \text{len}(e')}$	$\frac{\text{D-LET}}{\text{let}(e_1; x. e_2) \mapsto e_2[e_1/x]}$	

Figure 4: Substitution for the little language of numbers and strings

$$\begin{aligned}
 z[e/x] &\stackrel{\text{def}}{=} \begin{cases} e & \text{if } z \equiv x \\ z & \text{if } z \not\equiv x \end{cases} & \text{let}(e_1; y. e_2)[e/x] &\stackrel{\text{def}}{=} \text{let}(e_1[e/x]; y. e_2[e/x]) \\
 (\text{num}[n])[e/x] &\stackrel{\text{def}}{=} \text{num}[n] & \text{plus}(e_1; e_2)[e/x] &\stackrel{\text{def}}{=} \text{plus}(e_1[e/x]; e_2[e/x]) \\
 (\text{str}[s])[e/x] &\stackrel{\text{def}}{=} \text{str}[s] & \text{cat}(e_1; e_2)[e/x] &\stackrel{\text{def}}{=} \text{cat}(e_1[e/x]; e_2[e/x])
 \end{aligned}$$

Figure 5: Statics of the constants-and-functions fragment of the simply-typed λ -calculus

VAR $\frac{}{\Gamma, x : \sigma \vdash x : \sigma}$	NUM $\frac{n \in \mathbb{N}}{\Gamma \vdash \text{num}[n] : \text{Num}}$	PLUS $\frac{\Gamma \vdash e_1 : \text{Num} \quad \Gamma \vdash e_2 : \text{Num}}{\Gamma \vdash \text{plus}(e_1; e_2) : \text{Num}}$
TIMES $\frac{\Gamma \vdash e_1 : \text{Num} \quad \Gamma \vdash e_2 : \text{Num}}{\Gamma \vdash \text{times}(e_1; e_2) : \text{Num}}$	LET $\frac{\Gamma \vdash e_1 : \sigma_1 \quad \Gamma, x : \sigma_1 \vdash e_2 : \sigma_2}{\Gamma \vdash \text{let}(e_1; x. e_2) : \sigma_2}$	
LAM $\frac{\Gamma, x : \sigma \vdash e : \tau}{\Gamma \vdash \lambda x : \sigma. e : \sigma \rightarrow \tau}$	APP $\frac{\Gamma \vdash e_1 : \sigma \rightarrow \tau \quad \Gamma \vdash e_2 : \sigma}{\Gamma \vdash e_1(e_2) : \tau}$	

Figure 6: Dynamics of the constants-and-functions fragment of the simply-typed λ -calculus

VAL-LAM $\frac{}{\lambda x : \tau. e \text{ val}}$	VAL-NUM $\frac{n \in \mathbb{N}}{\text{num}[n] \text{ val}}$	D-PLUS $\frac{n_1 + n_2 = n}{\text{plus}(\text{num}[n_1]; \text{num}[n_2]) \mapsto \text{num}[n]}$
D-PLUS-1 $\frac{e_1 \mapsto e'_1}{\text{plus}(e_1; e_2) \mapsto \text{plus}(e'_1; e_2)}$	D-PLUS-2 $\frac{e_1 \text{ val} \quad e_2 \mapsto e'_2}{\text{plus}(e_1; e_2) \mapsto \text{plus}(e_1; e'_2)}$	
D-LET $\frac{}{\text{let}(e_1; x. e_2) \mapsto e_2[e_1/x]}$	D-APP-1 $\frac{e_1 \mapsto e'_1}{e_1(e_2) \mapsto e'_1(e_2)}$	D-BETA $\frac{}{(\lambda x : \tau. e_1)(e_2) \mapsto e_1[e_2/x]}$